

JIA-HUI PAN

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Introduction

I am a final-year Ph.D. student at the Department of Computer Science and Engineering of the Chinese University of Hong Kong (CUHK). My research primarily focuses on robotic manipulation. I am fortunate to be advised by Prof. Chi-Wing FU and to collaborate with Prof. Yun-Hui Liu through the Hong Kong Centre for Logistics Robotics. I also have the privilege of working remotely with Prof. Mingyu Ding from the University of North Carolina at Chapel Hill. I received my Master's degree and Bachelor's degree from Sun Yat-sen University (SYSU), where I was supervised by Prof. Wei-Shi Zheng and conducted research on video-based human action assessment.

Education

The Chinese University of Hong Kong
Ph.D. in Computer Science

(Aug. 2021 - Present)
Supervisor: Prof. Chi-Wing Fu

Sun Yat-sen University
M.Phil. in Computer Science

(Aug. 2018 - Jul. 2021)
Supervisor: Prof. Wei-Shi Zheng

Sun Yat-sen University
B.Sc. in Computer Science

(Aug. 2014 - Jun. 2018)
GPA: 3.8 out of 4.0

Publications

First-Author Papers

- * **Jia-Hui Pan**, Zhengzhe Liu, Ka-Hei Hui, Kai Chen, Xiaojie Gao, Pheng-Ann Heng, Qi Dou, Yun-Hui Liu, Mingyu Ding, and Chi-Wing Fu. "Human-Intuition-Driven Tree Packing for Diverse Everyday Objects." *In preparation for journal submission.*
- * **Jia-Hui Pan**, Yeok Tatt Cheah, Zhengzhe Liu, Ka-Hei Hui, Xiaojie Gao, Pheng-Ann Heng, Yun-Hui Liu, and Chi-Wing Fu. "OPA-Pack: Object-Property-Aware Robotic Bin Packing." *Conditionally accepted to IEEE Transactions on Robotics (T-RO).*
- * **Jia-Hui Pan**, Xiaojie Gao, Ka-Hei Hui, Shize Zhu, Yun-Hui Liu, Pheng-Ann Heng, and Chi-Wing Fu. "PPN-Pack: Placement Proposal Network for Efficient Robotic Bin Packing." *IEEE Robotics and Automation Letters (RA-L), 2024.*
- * **Jia-Hui Pan**, Ka-Hei Hui, Xiaojie Gao, Shize Zhu, Yun-Hui Liu, Pheng-Ann Heng, and Chi-Wing Fu. "SDF-Pack: Towards Compact Bin Packing with Signed-Distance-Field Minimization." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.*

- * **Jia-Hui Pan**, Jibin Gao, and Wei-Shi Zheng. "Adaptive Action Assessment." *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2021.
- * **Jia-Hui Pan**, Jibin Gao, and Wei-Shi Zheng. "Action Assessment by Joint Relation Graphs." *IEEE International Conference on Computer Vision (ICCV)*, 2019. (Oral)

Collaborative Papers

- Kai Chen, Chengkun Li, Chang Tu, **Jia-Hui Pan**, Yiyao Ma, Wei Chen, Zhongxiang Zhou, Xuecheng Xu, Stephen James, Chi-Wing Fu, Rong Xiong, Pieter Abbeel, Yun-Hui Liu, and Qi Dou. "A Retrieval-Augmented Framework Enabling VLM Spatial Awareness for Object-Centric Robot Manipulation." *Science Robotics*, 2026.
- Weiliang Tang, Jialin Gao, **Jia-Hui Pan**, Gang Wang, Li Erran Li, Yun-Hui Liu, Mingyu Ding, Pheng-Ann Heng, and Chi-Wing Fu. "Rethinking Intermediate Representation for VLM-Based Robot Manipulation." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- Rui Cao, **Jia-Hui Pan**, Yin Zhang, Qiwei Meng, Jiapeng Li, Gang Wang, Runsong Zhu, Wei Chen, Chi-Wing Fu, Jiangliu Wang, Yun-Hui Liu. "UA-Depth: Uncertainty-Aware Depth Restoration for Material-Agnostic Object Manipulation." *Under review at IEEE Transactions on Robotics (T-RO)*.
- Weiliang Tang, Dong Jing, **Jia-Hui Pan**, Zhiwu Lu, Yun-Hui Liu, Li Erran Li, Mingyu Ding, and Chi-Wing Fu. "Incentivizing Multimodal Reasoning in Large Models for Direct Robot Manipulation." *arXiv preprint arXiv:2505.12744*, 2025.
- Weiliang Tang, **Jia-Hui Pan**, Wei Zhan, Jianshu Zhou, Huaxiu Yao, Yun-Hui Liu, Masayoshi Tomizuka, Mingyu Ding, and Chi-Wing Fu. "Embodiment-Agnostic Action Planning via Object-Part Scene Flow." *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- Hao Feng, Zhi Zuo, **Jia-Hui Pan**, Ka-Hei Hui, Qi Dou, Jingyu Hu, Zhengzhe Liu. "Wonderverse: Extendable 3d scene generation with video generative models." *arXiv preprint arXiv:2503.09160*, 2025.
- Weiliang Tang, **Jia-Hui Pan**, Yun-Hui Liu, Masayoshi Tomizuka, Li Erran Li, Chi-Wing Fu, and Mingyu Ding. "GeoManip: Geometric Constraints as General Interfaces for Robot Manipulation." *arXiv preprint arXiv:2501.09783*, 2025.
- Jibin Gao, **Jia-Hui Pan**, Shao-Jie Zhang, and Wei-Shi Zheng. "Automatic Modeling for Interactive Action Assessment." *International Journal of Computer Vision (IJCV)*, 2023.
- Shao-Jie Zhang, **Jia-Hui Pan**, Jibin Gao, and Wei-Shi Zheng. "Adaptive Stage-Aware Assessment Skill Transfer for Skill Determination." *IEEE Transactions on Multimedia (TMM)*, 2023.
- Shao-Jie Zhang, **Jia-Hui Pan**, Jibin Gao, and Wei-Shi Zheng. "Semi-Supervised Action Quality Assessment with Self-Supervised Segment Feature Recovery." *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, 2022.

- Jibin Gao, Wei-Shi Zheng, **Jia-Hui Pan**, Chengying Gao, Yaowei Wang, Wei Zeng, and Jianhuang Lai. "An Asymmetric Modeling for Action Assessment." *European Conference on Computer Vision (ECCV)*, 2020.
- Jianfang Hu, Wei-Shi Zheng, **Jia-Hui Pan**, Jianguo Zhang, and Jianhuang Lai. "Deep Bilinear Learning for RGB-D Action Recognition." *European Conference on Computer Vision (ECCV)*, 2018.

Projects

- **Human-Instruction-Adaptive Robotic Bin Packing for Everyday Objects.**
(Apr. 2026 – Ongoing)
Role: *Team Leader*.
Developing a local large language model to capture diverse human packing preferences and enable adaptive robotic bin packing for everyday objects.
- **Object-Property-Aware Robotic Bin Packing for Everyday Objects.**
(May 2024 – Nov. 2024)
Role: *Team Leader*.
Leveraged large language models and deep Q-networks to develop an object-property-aware robotic bin packing framework.
- **Efficient General Object Packing via Placement Proposal Network.**
(Aug. 2023 – Dec. 2023)
Role: *Team Leader*.
Proposed a placement proposal network to efficiently generate near-optimal placements, significantly accelerating placement search for irregularly shaped supermarket objects.
- **A Pick-and-Pack System with a Franka Emika Robot Arm.**
(Mar. 2023 – Jul. 2023)
Role: *Team Leader*.
Developed a pick-and-pack system for irregularly shaped supermarket objects using instance segmentation and packing heuristics.
- **Gait Analysis for Stroke Patients.**
(Nov. 2017 – Mar. 2018)
Role: *Team Leader*.
Designed a regression-based model to learn healthy gait curve distributions for visualization and stroke gait pattern recognition.

Skills

- **Programming Languages:** most experienced with Python; experienced with L^AT_EX, Matlab, C and C++.
- **Languages:** Cantonese (native), Mandarin (native) and English (fluent).
- **Development Platforms and Softwares:** PyTorch, PyBullet, and TensorFlow.

May 30, 2026